

Amlogic_NPU_Support_Operator_List

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Revision history

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V0.1 (2026-07-15)	This is the First Version.

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1. Document Introduction

1.1 Document Overview

This document systematically sorts out and fully introduces the **operator set supported by the Amlogic NPU platform**, covering two categories: hardware-native accelerated operators and software-adapted supported operators.

All operators listed in this document strictly comply with the TFLite standard operator specification. For operators from other open-source frameworks such as ONNX and PyTorch, as long as they can be directly mapped, or fully match the operator set listed in this document after splitting, the original framework operators can run normally on the Amlogic NPU platform.

1.2 Applicable Platforms

This guide applies to the following Amlogic NPU platforms:

- C308L / C302X
- S928X
- A311D2
- T968D4
- S905X5 / S905D5
- C302X2
- C305X2/A123X
- A311Y3

Note: C305X2/A123X and A311Y3 belong to **ADLA3**, while other platforms belong to **ADLA2**.

1.3 Key Terms

- **ADLA2 (2nd Generation Amlogic NPU Chip):** The hardware instruction set only supports fixed-point integer formats such as int8, uint8, and int16, with no native floating-point acceleration unit.
- **ADLA3 (3rd Generation Amlogic NPU Chip):** Builds upon fixed-point computing capabilities by adding native hardware support for fp16 and bf16 floating-point formats, as well as int4 low-bit formats.
- **HW: Hardware Op,** executed on the NPU.
- **SW: Software Op,** executed on the CPU.

2. Operator List

2.1 ADLA2 Support List

2.1.1 Hardware Op List

Others: Currently, three data types including float32, int32 and int64 are supported. Some operators may only be compatible with one of the formats. For scenarios where a certain type is not supported temporarily, the corresponding type support can be added through subsequent iterations.

Operators	Int8/Uint8	Int16	Others	Note
ADD	HW	HW	SW	
AVERAGE_POOL_2D	HW	HW	SW	
CONCATENATION	HW	HW	SW	
CONV_2D	HW	HW	/	
DEPTHWISE_CONV_2D	HW	HW	/	
DEPTH_TO_SPACE	HW	HW	SW	
EMBEDDING_LOOKUP	HW	HW	SW	SW if the lookup tensor is a variable; HW otherwise.
FULLY_CONNECTED	HW	HW	/	
L2_POOL_2D	HW	HW	SW	
LOGISTIC	HW	SW	SW	
LSTM	HW	HW	/	
MAX_POOL_2D	HW	HW	SW	
MUL	HW	HW	SW	
RELU	HW	HW	SW	
RELU_N1_TO_1	HW	HW	SW	
RELU6	HW	HW	SW	
RESHAPE	HW	HW	SW	
RESIZE_BILINEAR	HW/SW	HW/SW	SW	HW if 1) input width*height is 1x1, or 2) resizing from [n, h, w, c] to [n, 2h, 2w, c] SW otherwise
RNN	HW	HW	/	
SPACE_TO_DEPTH	HW	HW	SW	
TANH	HW	SW	SW	

Operators	Int8/Uint8	Int16	Ohters	Note
PAD	HW	HW	SW	
GATHER	HW	HW	SW	SW if the lookup tensor is a variable; HW otherwise
BATCH_TO_SPACE_ND	HW	HW	SW	
SPACE_TO_BATCH_ND	HW	HW	SW	
TRANSPOSE	HW	HW	SW	
MEAN	HW	HW	SW	
SUB	HW	HW	SW	
SQUEEZE	HW	HW	SW	
UNIDIRECTIONAL_SEQUENCE_LSTM	HW	HW	/	
STRIDED_SLICE	HW	HW	SW	
SPLIT	HW	HW	SW	
BIDIRECTIONAL_SEQUENCE_LSTM	HW	HW	/	
PRELU	HW	HW	SW	
MAXIMUM	HW	HW	SW	
MINIMUM	HW	HW	SW	
LESS	HW	HW	SW	
NEG	HW	HW	SW	
PADV2	HW	HW	SW	
GREATER	HW	HW	SW	
GREATER_EQUAL	HW	HW	SW	
LESS_EQUAL	HW	HW	SW	
SELECT	HW	HW	SW	
SLICE	HW	HW	SW	
TRANSPOSE_CONV	HW	HW	/	
TILE	HW	HW	SW	
EXPAND_DIMS	HW	HW	SW	
EQUAL	HW	HW	SW	
NOT_EQUAL	HW	HW	SW	
SUM	HW	HW	SW	
SHAPE	HW	HW	SW	
REDUCE_MAX	HW	HW	SW	
PACK	HW	HW	SW	

Operators	Int8/Uint8	Int16	Ohters	Note
LOGICAL_OR	HW	HW	SW	
LOGICAL_AND	HW	HW	SW	
LOGICAL_NOT	HW	HW	SW	
UNPACK	HW	HW	SW	
REDUCE_MIN	HW	HW	SW	
REDUCE_ANY	HW	HW	SW	
SQUARE	HW	HW	SW	
FILL	HW	HW	SW	
RESIZE_NEAREST_NEIGHBOR	HW	HW	SW	
LEAKY_RELU	HW	HW	SW	
SQUARED_DIFFERENCE	HW	HW	SW	
MIRROR_PAD	HW	HW	SW	
ABS	HW	HW	SW	
SPLIT_V	HW	HW	SW	
REVERSE_V2	HW	HW	SW	
ADD_N	HW	HW	SW	
GATHER_ND	HW	HW	SW	SW if the indices tensor is a variable; HW otherwise
QUANTIZE	HW/SW	SW	/	(int8 <--> uint8) hw, others sw
HARD_SWISH	HW	HW	SW	
SELECT_V2	HW	HW	SW	
BATCH_MATMUL	HW	HW/SW	/	The T968D4/S905X5 does not support int16*int16, so it needs to be converted to SW.
CUMSUM	HW	HW	SW	
BROADCAST_TO	HW	HW	SW	
REDUCE_ALL	HW	HW	SW	
BROADCAST_ARGS	HW	HW	SW	

2.1.2 Software OP List

All Type: This refers to all possible types that the operator can exist.

Operators	All Type
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Operators	All Type
DEQUANTIZE	SW
FLOOR	SW
L2_NORMALIZATION	SW
LOCAL_RESPONSE_NORMALIZATION	SW
SOFTMAX	SW
CUSTOM	SW
DIV	SW
EXP	SW
TOPK_V2	SW
LOG_SOFTMAX	SW
CAST	SW
ARG_MAX	SW
SIN	SW
LOG	SW
SQRT	SW
RSQRT	SW
POW	SW
ARG_MIN	SW
REDUCE_PROD	SW
ONE_HOT	SW
FLOOR_DIV	SW
FLOOR_MOD	SW
RANGE	SW
UNIQUE	SW
CEIL	SW
COS	SW
ELU	SW
ROUND	SW
NON_MAX_SUPPRESSION_V4	SW

Operators	All Type
NON_MAX_SUPPRESSION_V5	SW
SCATTER_ND	SW
MULTINOMIAL	SW
GELU	SW

2.1.3 Limitation

Operators	Int8/Uint8	Int16	Limitation
AVERAGE_POOL_2D	HW	HW	max filter size is 15x15 if output is not 1x1
MAX_POOL_2D	HW	HW	max filter size is 15x15 if output is not 1x1

2.2 ADLA3 Support List

2.2.1 Hardware Op List

Operators	Int8/Uint8	Int16	f16/bf16	Others	Note
ADD	HW	HW	HW		
AVERAGE_POOL_2D	HW	HW	HW		
CONCATENATION	HW	HW	HW		
CONV_2D	HW	HW	HW		
DEPTHWISE_CONV_2D	HW	HW	HW		
DEPTH_TO_SPACE	HW	HW	HW		
EMBEDDING_LOOKUP	HW	HW	HW		SW if the lookup tensor is a variable; HW otherwise. Note
FULLY_CONNECTED	HW	HW	HW		
L2_POOL_2D	HW	HW	HW		
LOGISTIC	HW	SW	HW		
LSTM	HW	HW	HW		
MAX_POOL_2D	HW	HW	HW		
MUL	HW	HW	HW		
RELU	HW	HW	HW		
RELU_N1_TO_1	HW	HW	HW		
RELU6	HW	HW	HW		
RESHAPE	HW	HW	HW		
RESIZE_BILINEAR	HW/SW	HW/SW	HW		HW if 1) input width*height is 1x1, or 2) resizing from [n, h, w, c] to [n, 2h, 2w, c] SW otherwise
RNN	HW	HW	HW		
SPACE_TO_DEPTH	HW	HW	HW		
TANH	HW	SW	HW		
PAD	HW	HW	HW		
GATHER	HW	HW	HW		SW if the lookup tensor is a variable; HW otherwise
BATCH_TO_SPACE_ND	HW	HW	HW		
SPACE_TO_BATCH_ND	HW	HW	HW		
TRANSPOSE	HW	HW	HW		
MEAN	HW	HW	HW		
SUB	HW	HW	HW		
SQUEEZE	HW	HW	HW		
UNIDIRECTIONAL_SEQUENCE_LSTM	HW	HW	HW		

Operators	Int8/Uint8	Int16	f16/bf16	Ohters	Note
STRIDED_SLICE	HW	HW	HW		
SPLIT	HW	HW	HW		
BIDIRECTIONAL_SEQUENCE_LSTM	HW	HW	HW		
PRELU	HW	HW	HW		
MAXIMUM	HW	HW	HW		
MINIMUM	HW	HW	HW		
LESS	HW	HW	HW		
NEG	HW	HW	HW		
PADV2	HW	HW	HW		
GREATER	HW	HW	HW		
GREATER_EQUAL	HW	HW	HW		
LESS_EQUAL	HW	HW	HW		
SELECT	HW	HW	HW		
SLICE	HW	HW	HW		
TRANSPOSE_CONV	HW	HW	HW		
TILE	HW	HW	HW		
EXPAND_DIMS	HW	HW	HW		
EQUAL	HW	HW	HW		
NOT_EQUAL	HW	HW	HW		
SUM	HW	HW	HW		
SHAPE	HW	HW	HW		
REDUCE_MAX	HW	HW	HW		
PACK	HW	HW	HW		
LOGICAL_OR	HW	HW	HW		
LOGICAL_AND	HW	HW	HW		
LOGICAL_NOT	HW	HW	HW		
UNPACK	HW	HW	HW		
REDUCE_MIN	HW	HW	HW		
REDUCE_ANY	HW	HW	HW		
SQUARE	HW	HW	HW		
FILL	HW	HW	HW		
RESIZE_NEAREST_NEIGHBOR	HW	HW	HW		
LEAKY_RELU	HW	HW	HW		
SQUARED_DIFFERENCE	HW	HW	HW		
MIRROR_PAD	HW	HW	HW		
ABS	HW	HW	HW		
SPLIT_V	HW	HW	HW		
REVERSE_V2	HW	HW	HW		
ADD_N	HW	HW	HW		

Operators	Int8/Uint8	Int16	f16/bf16	Ohters	Note
GATHER_ND	HW	HW	HW		SW if the indices tensor is a variable; HW otherwise
QUANTIZE	HW/SW	SW	SW		(int8 <--> uint8) hw, others sw
HARD_SWISH	HW	HW	SW		
SELECT_V2	HW	HW	HW		
BATCH_MATMUL	HW	HW/SW	HW		The T968D4/S905X5 does not support int16*int16, so it needs to be converted to SW.
CUMSUM	HW	HW	HW		
BROADCAST_TO	HW	HW	HW		
REDUCE_ALL	HW	HW	HW		
BROADCAST_ARGS	HW	HW	HW		
DEQUANTIZE	HW	HW	HW		
FLOOR	HW	HW	HW		
L2_NORMALIZATION	HW	HW	HW		
SOFTMAX	HW	HW	HW		
DIV	HW	HW	HW		
EXP	HW	HW	HW		
LOG_SOFTMAX	HW	HW	HW		
CAST	HW	HW	HW		
ARG_MAX	HW	HW	HW		
SIN	HW	HW	HW		
LOG	HW	HW	HW		
SQRT	HW	HW	HW		
RSQRT	HW	HW	HW		
POW	HW	HW	HW		
ARG_MIN	HW	HW	HW		
REDUCE_PROD	HW	HW	HW		
ONE_HOT	HW	HW	HW		
FLOOR_DIV	HW	HW	HW		
FLOOR_MOD	HW	HW	HW		
RANGE	HW	HW	HW		
CEIL	HW	HW	HW		
COS	HW	HW	HW		
ELU	HW	HW	HW		
ROUND	HW	HW	HW		
SCATTER_ND	HW	HW	HW		
GELU	HW	HW	HW		

2.2.2 Software OP List

All Type: 指该算子可能存在的所有type。

Operators	All Type
LOCAL_RESPONSE_NORMALIZATION	SW
CUSTOM	SW
TOPK_V2	SW
UNIQUE	SW
NON_MAX_SUPPRESSION_V4	SW
NON_MAX_SUPPRESSION_V5	SW
MULTINOMIAL	SW

2.2.3 Limitation

Operators	Int8/Uint8	Int16	Limitation
AVERAGE_POOL_2D	HW	HW	max filter size is 15x15 if output is not 1x1
MAX_POOL_2D	HW	HW	max filter size is 15x15 if output is not 1x1